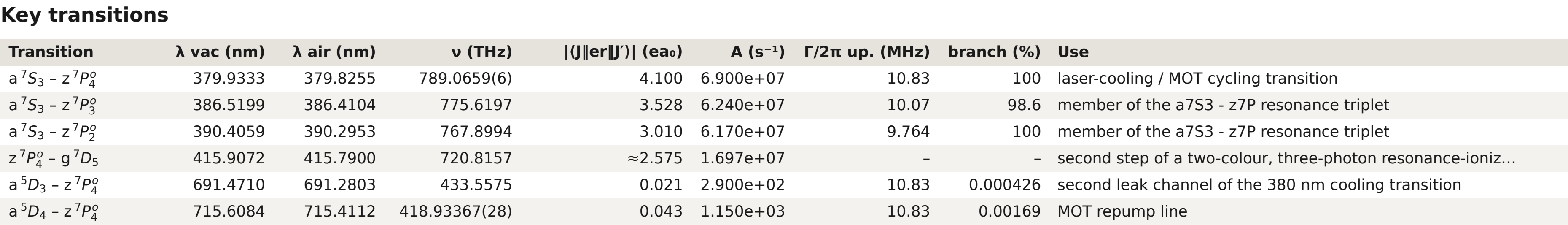
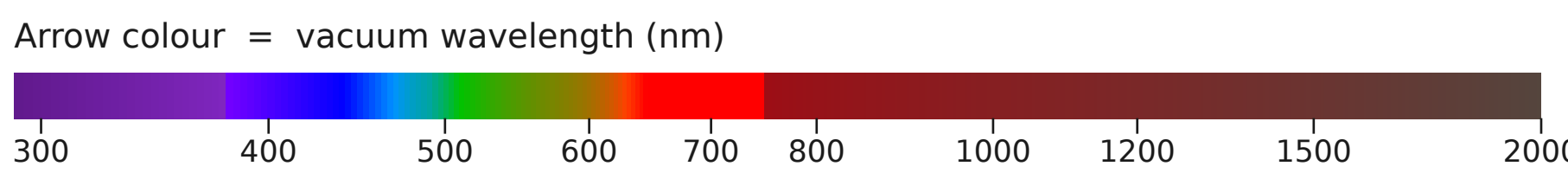


Mo (Mo I). Strongest classified transitions below 2 μm (NIST): 82 levels, 90 lines.

Mo⁺ ionisation limit 57204.30 cm⁻¹ = 7.09243 eV (174.812 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle U || e || J' \rangle$ in ea_0
 Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model or semi-empirical calculation.
 Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
 Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists)



⁹⁵Mo hyperfine structure (*I* = 5/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
<i>a</i> S ₃	-208.582060(10)	0.03705(10)	F=1/2: +2085.84 F=3/2: +1772.96 F=5/2: +1251.49 F=7/2: +521.44 F=9/2: -417.18 F=11/2: -1564.36

Measured A, B constants; sources in the data file.

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$a^7S_3 - z^7P_0^o$	379.933	92-96	920.4(19)
$a^7S_3 - z^7P_2^o$	379.933	94-96	418.4(16)
$a^7S_3 - z^7P_1^o$	379.933	98-96	-289.4(16)
$a^7S_3 - z^7P_4^o$	379.933	100-96	-812.0(19)